

Document Intelligence

AI Engineer Internship — Technical Case Study

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THE PROBLEM:

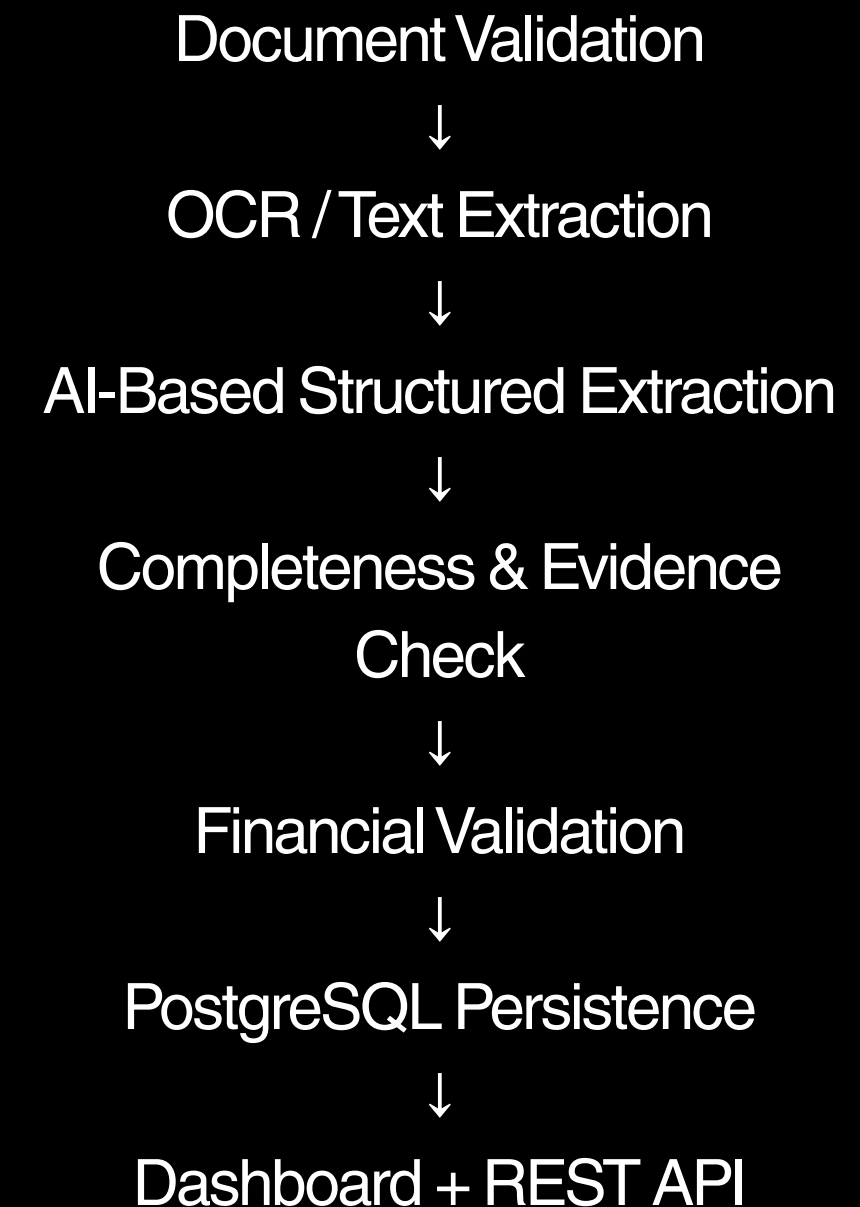
Financial documents contain:

- Unstructured text and complex tables
- Different layouts across documents
- Scanned/image-based content
- Comparative financial periods
- Numbers that must be mathematically consistent

THE SOLUTION:

Build an end-to-end system that converts these documents into structured, validated financial data.

Overview:



SUPPORTED DOCUMENTS

Invoice

- Vendor/customer
- Line items
- Quantity, price, discount
- Tax and totals

Balance Sheet

- Assets
- Liabilities
- Equity/capital
- Comparative periods

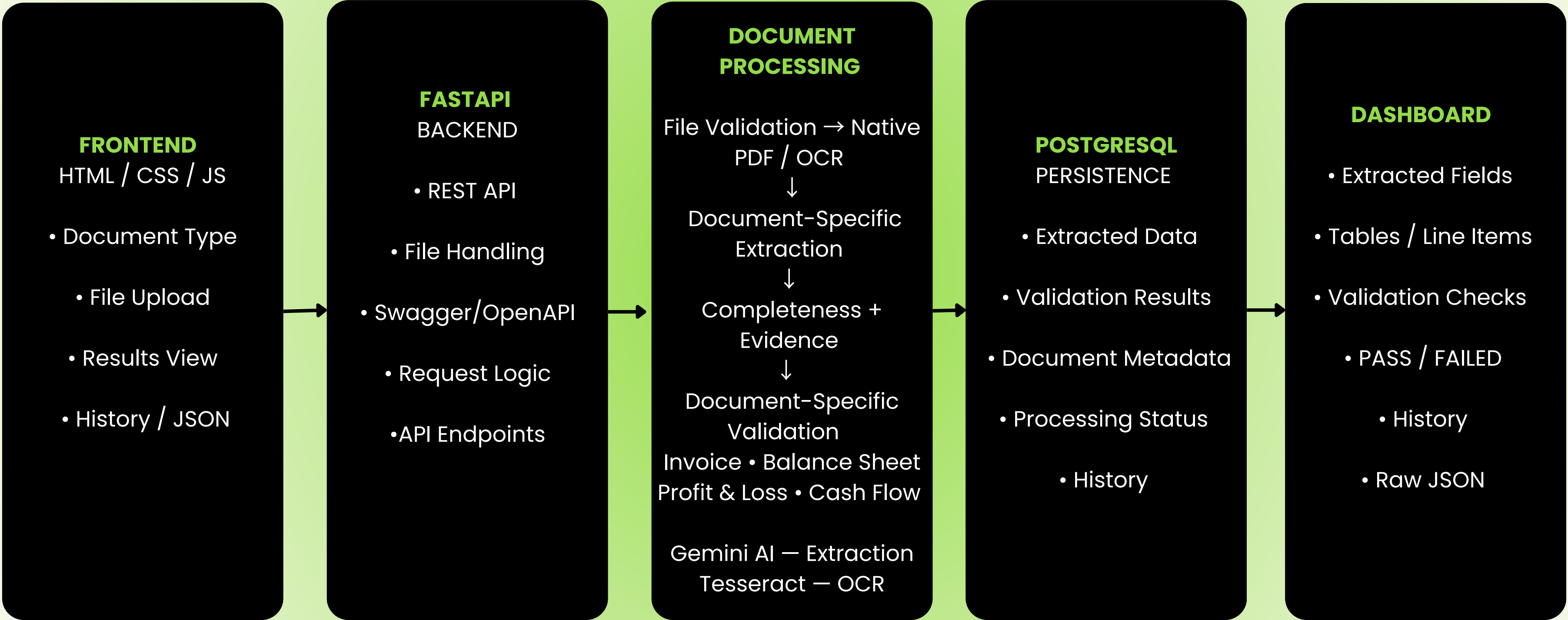
Profit & Loss

- Revenue/income
- Expenses
- Profit/loss
- Comparative periods

Cash Flow Statement

- Operating activities
- Investing activities
- Financing activities
- Opening/closing cash

SYSTEM ARCHITECTURE



DEPLOYED WITH RENDER

DOCUMENT PROCESSING PIPELINE

Step-by-Step Processing:

1. Validate

- File type
- Integrity
- Page count
- Readability

2. Extract Text

- Native PDF extraction
- OCR fallback for scanned documents

3. Understand

- Gemini-based document-specific extraction

4. Verify

- Completeness
- Source evidence
- Financial calculations

5. Store & Return

- PostgreSQL
- Structured JSON
- Processing status

INPUT VALIDATION

Validate Before Processing

The system rejects:

- Empty files
- Corrupted/unreadable files
- Unsupported formats
- Documents exceeding 3 pages

Supported Inputs:

- PDF
- JPG
- JPEG
- PNG

This prevents invalid documents from reaching the OCR/AI pipeline.

OCR & TEXT EXTRACTION

Handling Digital and Scanned Documents

Native PDF:

PDF → Native Text Extraction

Scanned PDF:

PDF → Page Images → Tesseract OCR → Text

Image:

Image → Tesseract OCR → Text

This allows the same extraction pipeline to work across both digital and image-based financial documents.

AI EXTRACTION

Gemini receives the document text and performs:

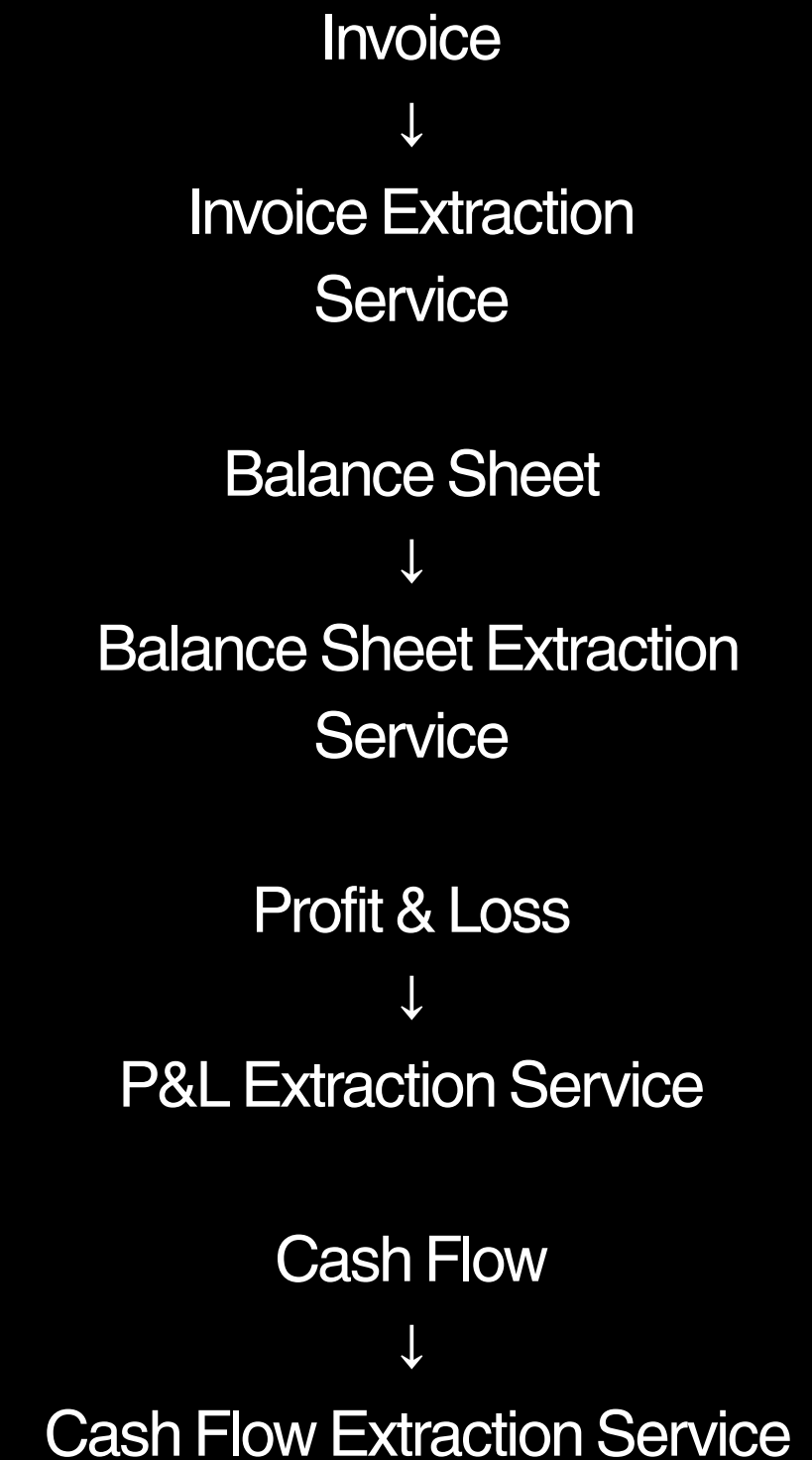
- Field extraction
- Table extraction
- Line-item extraction
- Comparative-period extraction
- Structured JSON generation
- Source evidence extraction

Why Document-Specific Services?

Different financial documents have different structures and validation rules. Dedicated services provide:

- Document-specific prompts
- Document-specific schemas
- Better table handling
- Easier validation
- Cleaner maintainability

Document-Specific Extraction



COMPLETENESS & EVIDENCE

Preventing Missing Information

After extraction, the system checks whether expected information has been captured.

The output can retain:

- Extracted value
- Source text
- Page number
- Missing values as null

Principle: Do not invent information that is not present in the source document.

FINANCIAL VALIDATION

Deterministic Validation Layer

AI extraction is followed by Python-based financial checks.
Every validation result contains:

- Formula / Check
- Inputs
- Calculated Value
- Reported Value
- Variance
- PASS / FAIL / NOT_APPLICABLE

This makes financial decisions transparent and reproducible.

DOCUMENT SPECIFIC VALIDATION

Invoice:

Quantity $\times$ Unit Price $\approx$ Line Total

Subtotal + Tax - Discount + Round Off $\approx$ Total

Balance Sheet:

Assets $\approx$ Liabilities + Equity

Profit & Loss:

Validate relationships between:

Income, Expenses and Profit

Cash Flow:

Operating + Investing + Financing + FX $\approx$ Net Change

Opening Cash + Net Change $\approx$ Closing Cash

API LAYER

REST API

Process and validate a document:

POST /api/v1/documents/process

View processing history:

GET /api/v1/documents

Retrieve the latest result:

GET /api/v1/documents/{document_name}

Service health check:

GET /api/v1/health

Swagger / OpenAPI

Interactive API documentation available through /docs.

STRUCTURED OUTPUT

One Common Response Structure:

Document

- └── Metadata
- └── Extracted Fields
- └── Tables / Line Items
- └── Evidence
- └── Financial Validation
- └── Processing Status

Example validation:

Check: Assets = Liabilities + Equity

Calculated: ₹10,00,000

Reported: ₹10,00,000

Variance: ₹0

Status: PASS

FRONTEND

User Workflow:

Select Document Type



Upload Document



Process



View Extracted Data



View Validation Results



View Raw JSON



Access Processing History

The dashboard makes both AI extraction and financial validation visible to the user.

PERSISTENCE

PostgreSQL

The system stores:

- Document metadata
- Document type
- Extracted structured data
- Validation results
- Processing status
- Processing metadata
- Document history

Reliability

PostgreSQL connection handling includes:

- pool_pre_ping
- pool_recycle

to reduce failures caused by stale database connections.

ERROR HANDLING & RELIABILITY

Failure Handling:

The system handles:

- Invalid uploads
- Corrupted documents
- OCR failures
- AI/API failures
- Database failures
- Validation failures

Result:

Instead of exposing internal errors, the API returns controlled error responses and meaningful processing status.

TESTING & DEPLOYMENT

Testing

24 automated tests passed

Coverage includes:

- File validation
- Financial calculations
- API processing flow
- Test database isolation

Live Verification

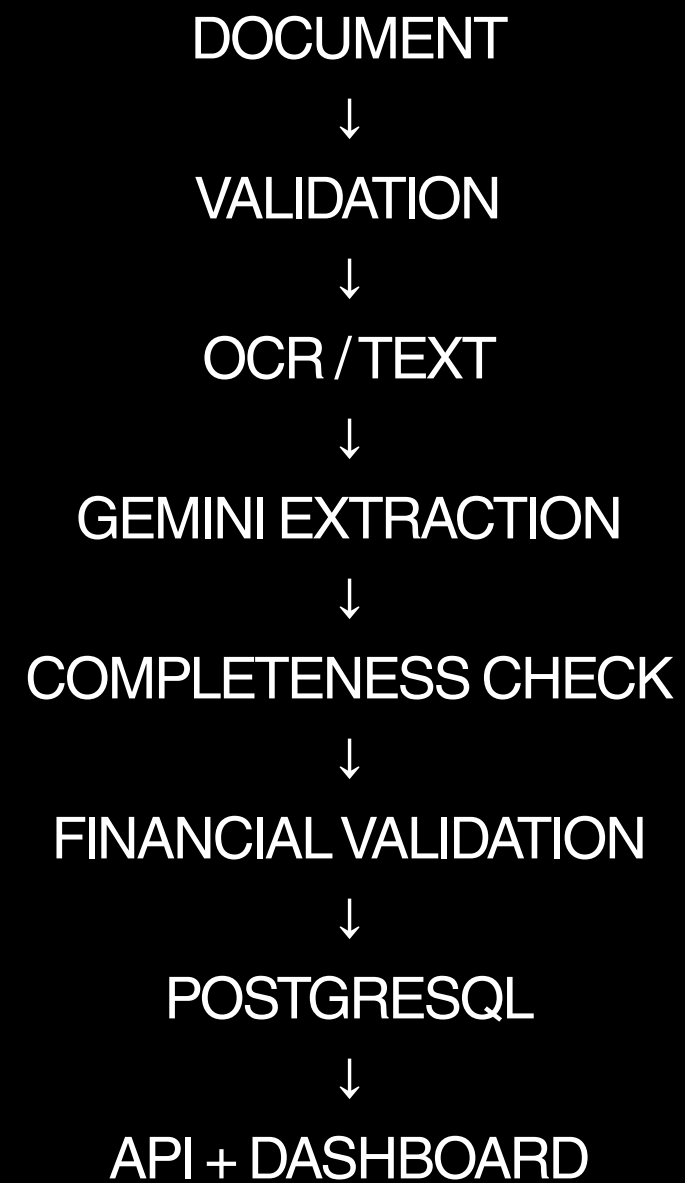
Successfully processed:

- Invoice
- Balance Sheet
- Profit & Loss
- Cash Flow Statement

Also verified corrupted PDF rejection through the deployed frontend.

FINAL SOLUTION

What We Built:
A complete financial Document
Intelligence platform



Final Outcome:

- Accurate structured extraction
- Explainable financial validation
- Persistent results
- Deployed application

From documents, we are able to extract trusted financial data.

Thank You